BSc in Artificial Intelligence

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Exam Project 3:

Customer Lifetime Value (CLTV) Analysis

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CLTV Forecast

Key Findings

R$ 575
Est. 90-day Customer Base Value

62 144
Total Customers Analysed

97.1%
One-Time Buyers

~3%
Avg. Cohort Return Rate

Hidden Trend Discovered

Retention is a structural problem, not a cohort-specific one. Monthly cohort analysis shows a uniformly low ~3% return rate across all acquisition periods (Oct 2016 – Mar 2018).
Implication: any retention initiative must actively intervene (post-purchase sequences, personalised discounts) — organic return behaviour is negligible.

Model Comparison

BTYD vs XGBoost

BG/NBD + Gamma-Gamma (BTYD)

- Interpretable probabilistic parameters
- Reliable aggregate revenue estimates per segment
- Best for: **Budget Forecasting & Cohort Planning**

XGBoost

- Top 10% targeting captures 2× more holdout revenue than BTYD
- Key drivers: Tenure (T), Total Spend, Frequency
- Best for: **Campaign Targeting & Personalisation**

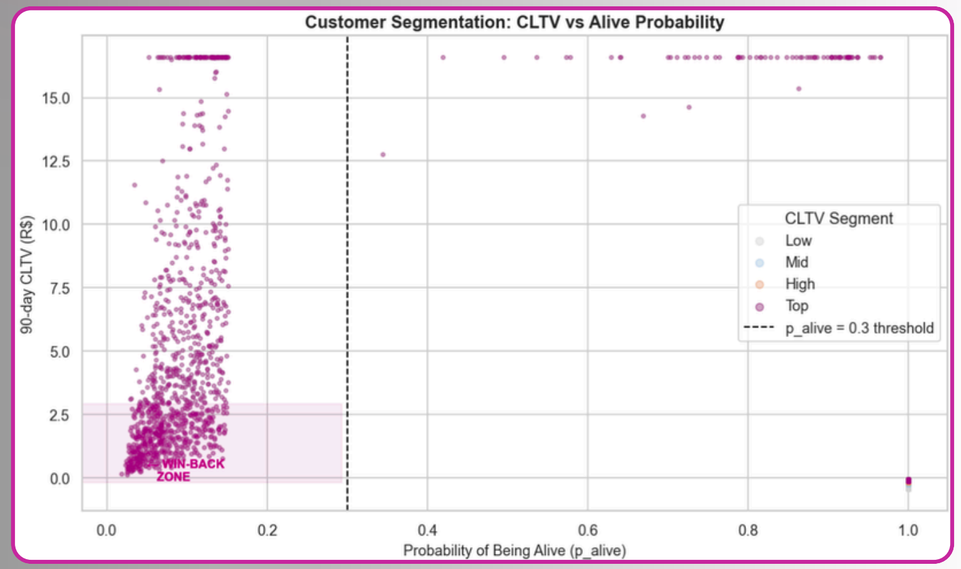
Key Gap

Neither model can reliably predict which one-time buyers will return — this is a data ceiling (97% one-time buyers), not a model failure. The two approaches are **complementary**: BTYD for *how much* to budget, XGBoost for *whom* to target.

Win-Back Campaign

3-Tier Targeting Strategy

Tier	Customers	Action	Channel	Goal
Highest Priority (Top Segment)	~200 high-value dormant	Personalised re-engagement: exclusive discounts, free shipping	Customer Service Call + VIP Email	Immediate Retention
Test & Learn (High Segment)	~850 mid-value dormant	Lower-cost outreach: product recommendations, targeted coupons	Email Campaign + Push Notification	Test Responsiveness
Retain & Grow (Top Segment)	Active high-value	Loyalty programme, upselling, frequency incentives	Email Newsletter + In-App	Nurture & Increase Frequency



Strategic Recommendation:

Focus the retention budget on the **1,048 win-back candidates** (High/Top CLTV, p_alive < 0.3) — estimated recoverable value: **R$ 6,061 over 90 days**.
Use **BTYD** to set the ***budget envelope*** and **XGBoost** to ***rank and prioritise*** individual customers within each tier. This dual-model approach maximises both forecasting accuracy and campaign ROI.